

Question Paper

Exam Date & Time: 24-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: MIE 1071/MIE_1071 BASIC MECHANICAL ENGINEERING

BASIC MECHANICAL ENGINEERING [MIE 1071]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Steam is generated in a boiler at a pressure of 10 bar. The steam leaving the boiler initially has a specific enthalpy of 2100 kJ/kg. It is then passed through a superheater, where it is further heated at the same pressure until its enthalpy reaches 2700 kJ/kg. (4)
- i) Determine the initial quality of the steam leaving the boiler and the quality after partial superheating.
- ii) Alternatively, if the steam needs to be further heated in the superheater to become superheated by 10°C, calculate the additional heat required per kilogram of steam from the boiler exit.

Course Outcome: "CO3" / Cognitive Level: "Remember"

- 1B) A steam boiler produces 6000kg of steam per hour at a pressure of 0.56 MPa and at a temperature of 230°C. The feed water temperature is 25°C. The mass of fuel consumption is 20 Kg/min., having a calorific value of 24MJ/kg. Assess the improvement in boiler efficiency if the following accessories are used. (3)
- i) Economizer will increase the feed water temperature to 63°C and reduce coal consumption by 10%.
- ii) Air Preheater which will reduce coal consumption by 22%.

Course Outcome: "CO3" / Cognitive Level: "Remember"

- 1C) Compare the functions of blow-off cock and fusible plug. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2A) A mechanical drive system in a mining plant uses a flat belt drive to transmit power between two parallel shafts rotating in opposite directions. The driver pulley has a diameter of 300 mm while the driven pulley on the output shaft runs at 250 rpm. The centre distance between the two pulleys is 800 mm. The initial belt tension is 1500 N, and the coefficient of friction between the belt and pulley surface is 0.28. (4)
- Determine the following:
- i) The tensions in the tight side and slack side of the belt.
- ii) The power transmitted by the belt.
- iii) The length of the belt required for this belt drive. Assume the system provides a speed reduction ratio of 3:2.

Course Outcome: "CO3" / Cognitive Level: "Remember"

- 2B) A compound gear train consists of six gears - A, B, C, D, E, and F. Gear A is the driver, and gear F is the driven gear, having 80 teeth and 18 teeth, respectively. Gear A meshes with gear B, and gears B and C are mounted on the same shaft. Gear C, which has 60 teeth, meshes with gear D. (3)

Gears D and E are mounted on another common shaft, and gear E finally meshes with gear F. The overall velocity ratio from gear A to gear F is 64. It is also given that the velocity ratio between A and B is equal to the velocity ratio between C and D, and equal to the velocity ratio between E and F. Determine:

- i) The number of teeth on gears B, D, and E.
- ii) The centre distance between the driver and driven shafts, if the circular pitch of all gears is 2.5π mm.

Course Outcome: "CO2" / Cognitive Level: "Remember"

- 2C) Explain the effect of creep and slip on the velocity ratio of belt drives. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Illustrating thermodynamic events and piston movements, explain the working of an IC engine fitted with a heavier flywheel and operating on isochoric heat addition principle. (4)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3B) The total clearance volume of a 6-cylinder internal combustion engine is 300cc. Determine the bore and stroke of each cylinder if the two are equal for each cylinder and the compression ratio is 10:1. Also calculate power output per liter of cylinder stroke volume if the total power output of the engine is 30 kW. (3)

Course Outcome: "CO1" / Cognitive Level: "Remember"

- 3C) What criteria are used to classify internal combustion engines? (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4A) A long, gradual external taper is to be produced on a mild steel shaft whose length exceeds the compound rest travel. Identify the most suitable taper turning method, justify your choice, and explain its working principle with a neat sketch. (4)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 4B) A component requires a smooth, accurately sized hole with internal threads to fit a conical head screw. (3)
List the sequence of machining operations that need to be performed on a drilling machine to obtain the required hole, and briefly state the purpose of each operation.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 4C) Two thin steel sheets used in automobile body fabrication are to be joined quickly without using filler material. Identify the welding process and explain the same. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 5A) Give reason for the following: (4)
i) Gate cutting is done to create a passage through which molten metal enters the mold cavity.
ii) Patterns are made slightly larger than the final casting dimensions.
iii) Molding sand must have sufficient cohesiveness and proper moisture content.
iv) Patterns are provided with a draft or taper.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5B) Explain the role of industrial robots in processing and material handling applications. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5C) Justify the following statements: (3)

- i) Automation helps increase labour productivity
- ii) Additive manufacturing helps increase customization in parts being manufactured
- iii) CNC machines reduces manufacturing lead time

Course Outcome: "CO3" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 MIE_1071:BASIC MECHANICAL ENGINEERING

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	19
3	3	CO 3	28
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 MIE_1071:BASIC MECHANICAL ENGINEERING

#	Level	No of Questions	Marks (%)
1	Remember	5	17(34%)
2	Understand	4	12(24%)
3	Application	2	6(12%)
4	Analysis	4	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)